

Cute Auto Karyotype ✨ User Manual

Version 1.0

1. Software Overview

Cute Auto Karyotype is a software application designed to assist with chromosome detection, cropping, editing, numbering, and arrangement from metaphase images for rapid karyotype board preparation. The software integrates essential cytogenetic image-processing steps into a single user-friendly interface.

The software is suitable for:

- Veterinary Cytogenetics
- Animal Genetics
- Conservation Genetics
- Research Laboratories
- Educational Purposes

2. Installation

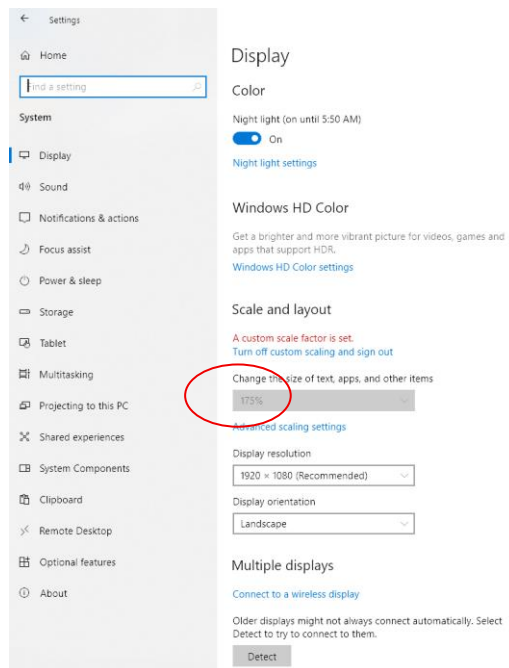
Method 1: Using the Precompiled Release (Recommended)

1. Download the software package.
2. Extract the ZIP file.
3. Double-click:

RUN_Cute_Karyotype.bat

4. The software is ready to use.

Note: For proper display of the graphical user interface, Windows Display Scaling is recommended to be set between 100%–175%



3. Recommended Workflow

Step 1: Load an Image

Click:

Load Image

Select a metaphase image for analysis.

Supported formats:

- **JPG**
- **PNG**
- **BMP**
- **TIFF**

Step 2: Detect Chromosomes

Click: [Detect] The software automatically identifies candidate chromosome regions within the metaphase image.

Step 3: Review Detection Results

Click each detected box and choose: Keep to retain the chromosome candidate, Or [Reject] to remove it from further analysis. If a chromosome is not detected automatically, use: Add Box to manually add a chromosome region.

Step 4: Remove Rejected Candidates

Click: [Delete Rejected] to permanently remove all rejected boxes from the workspace.

Step 5: Automatic Chromosome Numbering

Click: [Auto Number] to automatically assign chromosome numbers according to the selected sorting order.

Note

If chromosomes have been edited in the Crop Field and Auto Number is applied again, previously removed artifacts may reappear.

Step 6: Edit Chromosome Images

Select a chromosome in “Crop Field”. The selected chromosome will be displayed in “Active Crop Editor”

Available editing functions include:

- Removing artifacts
- Removing neighboring chromosome fragments
- Cleaning background debris
- Undo
- Redo

Step 7: Rotate Chromosomes

Chromosomes can be rotated using:

- Rotation buttons
- Custom angle input

Rotation Angle Convention

+ Angle

Rotates the chromosome counterclockwise.

- Angle

Rotates the chromosome clockwise.

Step 8: Export Results

Export the final karyotype as: PNG or PDF for reports, presentations, theses, and publications.

Output Files

[chromosome_crops/](#)

Contains chromosome images cropped directly from the original metaphase image.

[chromosome_layers_transparent/](#)

Contains each chromosome as an individual transparent PNG image.

These files can be imported into:

- Microsoft PowerPoint
- Adobe Photoshop

for chromosome rearrangement, annotation, and publication-quality figure preparation.

[karyotype_sheet.png](#)

A raster-image karyotype sheet generated automatically by the software.

[karyotype_sheet.pdf](#)

A PDF version of the final karyotype sheet suitable for printing and archiving.

[karyotype_sheet_editable_layers.svg](#)

An editable vector-based karyotype sheet.

Compatible with:

- Adobe Illustrator
- Inkscape

for advanced figure editing and publication preparation.

[numbered_original_overlay.png](#)

The original metaphase image with chromosome numbering overlaid.

Useful for:

- Verification of chromosome identification
- Traceability between chromosomes and their original positions in the metaphase spread

[review_table.csv](#)

A CSV file containing chromosome metadata, including:

- Chromosome ID

- Coordinates
- Bounding Box Dimensions
- Keep/Reject Status
- Assigned Chromosome Number

Step 9: Create the Karyotype Board

Click: [Create Final Board] to arrange all selected chromosomes into a karyotype sheet.

Within the Final Board Editor, users can:

- Drag chromosome positions
- Resize chromosomes
- Rearrange chromosome order
- Export the final karyotype as: PNG or PDF or PPT for reports, presentations, theses, and publications.

4. Troubleshooting

- The software has not started. Run: **INSTALL_Cute_Karyotype.bat** before launching the program.
- Some buttons or controls are missing. Check **Windows Display Scaling settings**. Recommended: 100–175%
- Some chromosomes are not detected. Use: **[Add Box]** to manually add missing chromosome regions.
- Excessive image artifacts are detected Use: **[Reject][Brush][Lasso]** to remove artifacts before creating the Final Board.

5. Additional Information

Cute Auto Karyotype 🌟

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